



NATIONAL INSTITUTE OF BUSINESS MANAGEMENT
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Internet of Things Course Work
Smart Pet Feeder

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ABSTRACT

The Smart Pet Feeder functions as a modern solution because it helps resolve pet owners' issues regarding proper feeding during periods when owners cannot be present. Through IoT-enabled automation this system delivers precise feeding schedules for pets including food and liquid distributions regardless of owner absence. The system enables remote monitoring and status oversight of feeders and schedule setups through its combination of smart schedules and internet controls.

Through its mobile application built with Flutter and backed by Firebase Realtime Database the Smart Pet Feeder delivers full voice control functionality for operating the system. Users of this automated feeder device control how their pets eat by setting feeding preferences from the mobile app in which they can observe device status and enable manual intervention whenever needed.

The feeder utilizes smart sensors to measure feeding amounts automatically thereby doing away with manual food controls. Pet owners can enjoy management ease along with reassurance that their pets receive proper feeding care because of this system despite being away.

Smart Pet Feeder serves both pet owners through its benefits while supporting developers who integrate IoT with pet care systems development. This innovative feeding system feeds pets efficiently using sophisticated methodology without complex user interfaces thus serving contemporary families who consider their pets' well-being utmost. This advanced system provides better pet feeding protection through its safe and reliable and efficient operation which surpasses conventional pet feeding solutions.

ACKNOWLEDGEMENT

Our thanks go first to Mr. Supun Asanga for directing our work on developing the IoT pet feeder system. He stands by us with both great solutions and ongoing support that helped us build out this project. Mr. What Asanga taught us helped us solve our difficulties and follow the right path to reach our success.

Our team is greatly thankful for his remarkable results which created motivational energy across our project. His dedication to producing error-free solutions and his motivating behavior kept us focused on building a strong practical system for everyone to use.

Our team denies its development team for their ongoing dedication to create structures that boost project performance. Our collaboration with these experts produced a pleasurable outcome for us. Our team extends deep thanks to Mr. Supun Asanga for enabling our work. Through his individual actions he made our project successful and his trust enabled us to turn our idea into reality. His leadership has been a privilege to work with but also a pleasure to experience. We are grateful for his backing.

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1. INTRODUCTION

The love connection between pet owners and their pets faces challenges because work responsibilities fill much of their day. The Smart Pet Feeder combines Internet of Things (IoT) technology to deliver pets their scheduled feedings with exact portion sizes. Smartphones and Firebase integration with the system creates real-time pet monitoring capabilities alongside flexible feeding routines and precise food dispensing which both simplifies owner life and enhances pet's health results.

1.1 Project Objectives

❖ Dynamic Feeding Based on Pet Activity

- The feeder uses motion sensors to measure pet activity. The device monitors pet movement to program automatic feedings at appropriate times for active meals. The system helps pets eat better because they will not receive excessive amounts of food or skip meals because of scheduled feeding times.

❖ Real-Time Inventory Monitoring

- Our weighing tools inside the food and water trays display real-time stock readings. Cloud alerts go to owners on their phones to control supplies before supplies reach critical levels so pets never run out and owners can deliver fresh supplies promptly.

❖ Enhanced Mobile App

- The system uses a simple app design combined with its cloud connection for convenient notifications. Through a connected app owner get real-time data about their pet feeders plus they can update settings and check feeder status even while away from home. Our system keeps pet feeding on schedule even if owners need to step away from home.

❖ **Energy-Efficient Design**

- Parts of the system use energy wisely so it keeps running with little support or power top-ups. The device runs efficiently on limited power which lowers operational costs.

❖ **Advanced Scheduling**

- The feeder enables users to design separate feeding schedules according to their preferences. The programmable settings help to serve each pet's individual dietary needs whatever their specific eating requirements.

1.2 Significance of the Project

The Smart Pet Feeder brings pet care convenience through solutions which resolve frequent feeding problems faced by pet owners. Key features of the system and their significance include:

❖ **Advanced Motion Detection:**

- The system uses pet habits to detect physical activity then modifies the meal times automatically.
- The device controls exactly how much food enters the feeder according to animal activities to make sure pets eat correctly.

❖ **Real-Time Alerts:**

- This technology protects pets from neglect and supplies food and water constantly as owners take care of business.

❖ **Comparison with Traditional Feeders:**

- In contrast to regular automatic feeders this IoT-connected device automatically adjusts feeding schedules and measurements based on how pets behave real-time. The system works with a phone app to enable users from anywhere to control pet care features.

2. METHODOLOGY

Our approach starts by finding system difficulties then builds a sensor-based device connected to cloud computing systems and ends with creating an easy-to-use mobile app on Flutter. Firebase handles data storage and notifications. Testing helps us check that our sensors work correctly and our system operates dependably. During deployment we make sure sensors connect well and adjust them to work their best.

❖ Requirement Analysis:

- Start by listing the distinct problems pet owners encounter which include handling meal schedules, preventing calorie excess, and tracking food and water supply in real time. Study how current systems function to spot missing elements and then engineer new solutions to solve these problems better.

❖ System Architecture Design:

- Create a total system design that includes motion detectors weight scales and cloud network processing. Depict the interactions of system parts through visual design which shows the connections between hardware, programming tools and user screens.

❖ Development Process:

- Build a complete mobile app with Flutter by providing easy navigation and powerful features. Create a Realtime Database in Firebase to manage your data well and let Firebase Cloud Messaging give instant updates. Attach motion and weight sensors directly to program hardware to make it operate as one system.

❖ Testing and Validation:

- Test our software against real-world scenarios both for motion detection and inventory monitoring operations. Test how well cloud notifications deliver messages in different network conditions and adjust system reactions to maintain best performance.

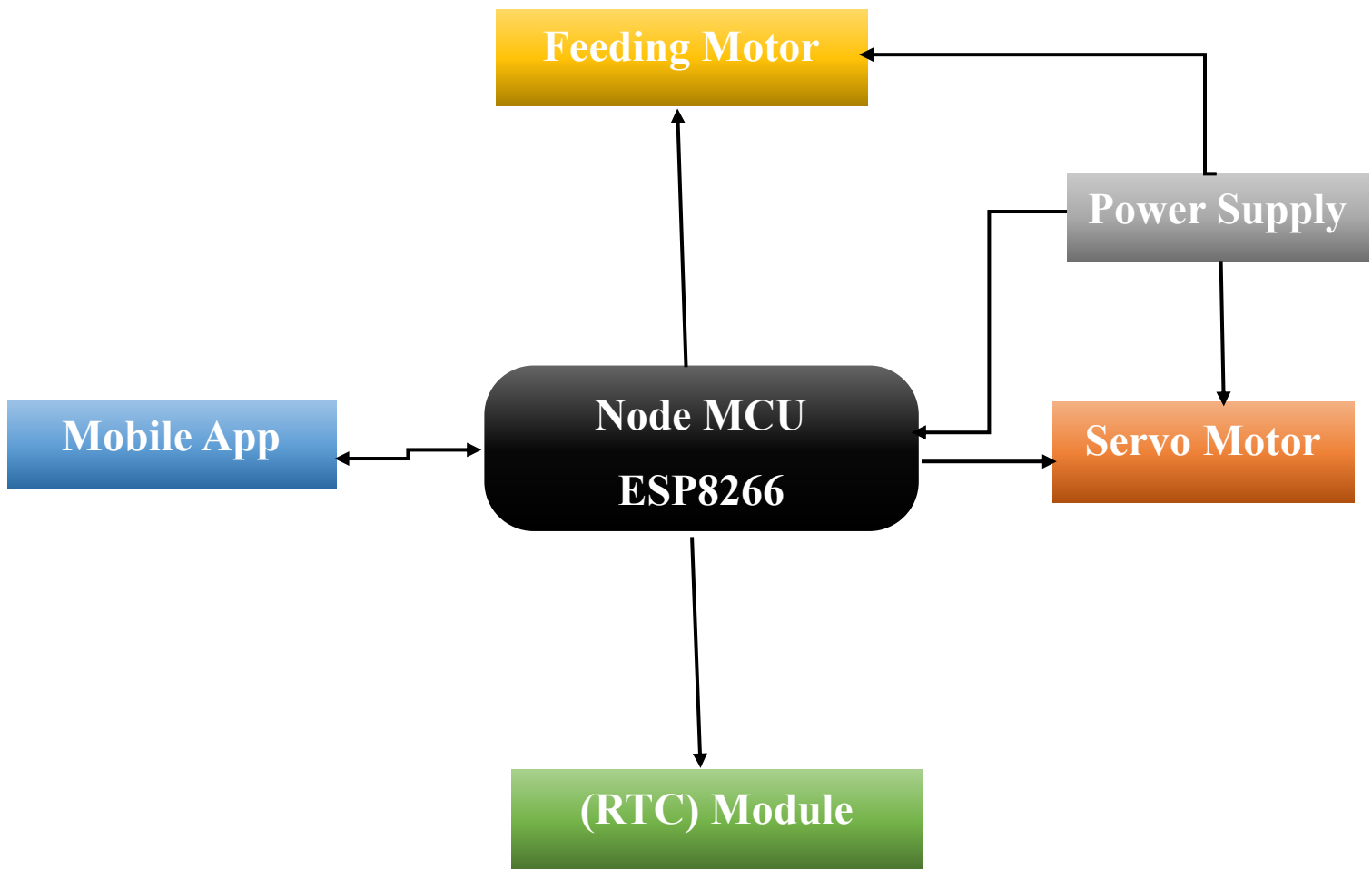
❖ **Implementation and Deployment:**

- Test sensors carefully to deliver exact measurement results. Test how different parts work together after installation and then test the system at its real-world setting. Implement the system for service and check that each element collaborates without problems.

3. SYSTEM DESIGN AND ARCHITECTURE

The system architecture includes several key components:

- ❖ **NodeMCU ESP8266:** The heart of the system, enabling Wi-Fi communication.
- ❖ **Servo Motor:** The device outputs food servings with exact measurements following input from users.
- ❖ **Water Motor:** The water motor program of a smart pet feeder controls automated water distribution at preset times while providing manual and automated control through an app interface.
- ❖ **Food Container:** Maintains a stock of sufficient capacity to be able to feed everybody for several days.
- ❖ **Relay Module:** Through its switching operation this device controls the food and water dispensing motors for automatic feeding and watering.
- ❖ **Breadboard Power Supply:** A stable regulated voltage supply feeds the circuit components to guarantee correct function for the microcontroller sensors and motors.
- ❖ **Mobile Application:** The developers created this system using the Flutter framework for distant management capabilities.
- ❖ **Firebase Realtime Database:** An integrated storing system and database engine function to automatically manage feeding information while keeping user data accurately synchronized.



4. IMPLEMENTATION

The implementation of the Smart Pet Feeder involves the following critical stages:

4.1 Hardware Setup

To configure the pet feeder system the hardware execution requires building its physical mechanical elements. The main hardware components are:

- **NodeMCU ESP8266:** The device functions as the operating center by linking WiFi capabilities with feeder management tasks. The technology enables live communication between mobile application interfaces and physical hardware controls which regulates real-time feeding operations.

- **Servo Motor:** The dispensing mechanism operates under the control of the servo motor. This equipment rotates the food dispenser compartment in order to deliver pre-measured food amounts. The servo motor requires exact control capabilities to deliver proper portions of food. The servo motor achieves precise measurement by maintaining calibration during dispensing operations.
- **Water Motor** A smart pet feeder that uses a water motor functions as a vital component for delivering continuous clean water to pets. The automatic water dispenser operates according to preestablished timers to maintain fresh water access and guard against water stagnation for pets. Remote water dispensing control is possible through mobile app programming whereby pet owners can select manual operation or automatic scheduling for the motor. Water motor-equipped advanced feeders possess sensors which help identify depleted water levels so they can automatically refill the pet bowl with clean water supply.
- **Food Container:** The storage mechanism retains food inside until the machine delivers it. With its proper capacity the container can store enough food that matches the feeding requirements of the pet for a period of several days.
- **Relay Module:** Through controlling operations of food and water dispensing motors the relay module functions as an electronic switch. The device gives the microcontroller capability to control powerful devices through a low voltage command. The relay module turns the power on to the motors through the completed circuit thus allowing them to distribute food and water at their programmed times. The automated dispensing function paired with this module enables automated feeding and watering of pets without manual contact therefore securing a consistent pet feeding schedule.

- **Breadboard Power Supply:** A stable regulated voltage stream from the power supply unit maintains correct operation for the circuit electronics together with the microcontroller sensors and motors. The power supply unit transforms incoming voltage into appropriate power levels to stop potential operational failures. When integrated with a dependable PSU system operation stability along with performance quality suffers significant improvement.

4.2 Mobile Application (Flutter Development)

A popular cross-platform application programming framework named Flutter powers the mobile app development. Through this application users establish their main communication channel with the Smart Pet Feeder. Key features of the mobile application include:

- **Feeding Schedule Setup:** Through their mobile application users can establish meal schedules because they define both the start times and measurements of their pet's food portions. Through pre-set schedules owners can schedule their pet's meals to automatically feed even when they are absent from home.
- **Remote Control:** Users maintain pet feeder control through the application's remote features. Users gain complete control through a mobile interface which allows manual triggering of feeding functions or direct oversight of feeding schedule times and portion amounts.
- **Real-Time Monitoring:** Using Firebase integration ensures real-time data communication between mobile app and pet feeder operation. Users gain visibility into feeder status while simultaneously viewing current container contents alongside the capability to carry out needed adjustments.
- **User Interface:** The application follows straightforward design principles which make it possible for users at every tech level to effortlessly use and control the system. Through this interface users establish schedules and observe pet feeding performance and they can manually turn on feeding sessions.

4.3 Firebase Integration for Real-Time Data Storage and Synchronization

Through Firebase integration the mobile app maintains real-time data connectivity with the pet feeder hardware. The mobile app utilizes Firebase Realtime Database to manage the storage of feeding schedules and pet information together with user preferences.

This integration ensures:

- **Real-Time Updates:** System-wide data changes initiated from the app automatically appear throughout the platform for synchronized operations between mobile applications and directive hardware.
- **Data Security:** With Firebase users enjoy complete protection of feeding schedules and pet details because the system permits access only to authorized users.
- **Data Access:** The system operates efficiently through data synchronization because mobile users and the feeder component both access the Firebase database to obtain and modify schedules, dispensing parameters as well as user preference settings.

5. TOOLS AND TECHNOLOGIES USED

5.1 Hardware:

- NodeMCU ESP8266
- Servo Motor
- Water Motor
- Food Container
- Relay Module
- Breadboard Power Supply

5.2 Software:

Flutter (Mobile Application Development)

Firebase Realtime Database which is used for Data Storage.

Arduino IDE

6. RESULTS

The Automated Pet Feeder operated successfully through three key functionalities.

○ **Scheduled and Accurate Food Dispensing:**

The device delivers food according to the established time schedule which users predetermined. The pet feeding mechanisms built with Real-Time Clock modules perform precise schedule functions which prevents pets from going hungry due to missed portions.

○ **Seamless Remote Control and Monitoring:**

The built Flutter-based pet mobile application enables pet owners to use an interface for controlling their feeder. The system provides capabilities to manage a set schedule with feeding operations and it enables switching automatically between automatic and manual modes according to present activities.

○ **Reliable Data Storage and Retrieval:**

The system benefits from better security because Firebase Realtime Database serves as the storage location for feeding schedules and user preferences and system configurations. The system maintains real-time data accessibility for both the feeding device and related mobile application features from start to finish.

○ **Energy-Efficient Operation:**

The NodeMCU chip along with the motors possess low power consumption capabilities as part of their hardware design specifications. The system operates indefinitely by charging less frequently when compared to those using LI-ion batteries.

○ **User-Friendly Interface:**

The mobile app stands as a new tool for users but the hardware interface along with the application maintain simple layouts which avoid frustrating the end-user. Any user who lacks technical expertise can install monitoring settings to track the system's condition and modify it effortlessly.

7. CHALLENGES AND LIMITATIONS

7.1 Challenges

- Ensuring accurate dosing required extensive calibration of the servo motor: The servo motor for controlling the food dispensing was another sensitive aspect that had to be tested severally to get the desired standards to fit the servo motor.
- Establishing stable Wi-Fi communication in areas with weak signals:

For real-time synchronization of the databases, the system depends massively on Wi-Fi connectivity. Equipment performance was also affected by limited connectivity in regions with low signal strength.

- Synchronizing data between the mobile app and Firebase in real time: Coordination of synchronous activity presented technological complexities particularly while establishing data coordination during low bandwidth.

7.2 Limitations

- System functionality depends on continuous internet connectivity: This means that the system is extremely dependent on having an internet connection for the remote-control commands and scheduling.
- Limited to dry food types due to the nature of the dispensing mechanism: It is crucial to provide a wide range of applications since different pets may need other types of foods, but the feeder is designed only for dry foods, and non-dry foods cannot be dispensed.

8. FUTURE IMPEMETATIONS

The upcoming Automated Pet Feeder project contains multiple enhancements to improve both the interface and the pet features. Additional details follow the description for each structural aspect:

○ The system adds a Camera Module for providing owners with live monitoring capabilities of their pets.

Users can monitor their pets' real-time condition whenever the camera module gets activated. The pet monitoring system enables owners to observe their pets while feeding to confirm nutritional consumption and it reduces food spillage. Also it lets owners check on pets when away from home. Users who have this device can talk to

their animals through the camera system even though such communication is an additional capability.

○ **Extending Support to Dispense Wet Food and Water:**

Depending on dry food diets the current system operates effectively though pets occasionally eat wet food products together with water in their diet. Extending the feeder's existing food capacity to allow wet food types should be pursued since this would benefit owners. A storage system together with distribution methods would need addition to handle wet foods since moisture ruins them naturally. Pet parents can benefit from integration of water dispensing features because their pets will receive water when the owner is not available to feed them.

○ **Adding Voice Assistant Integration for Hands-Free Control:**

The pet feeding device enables users to manage it through voice commands by integrating additional voice assistant solutions such as Google Assistant and Alexa as well as Siri. Using this addition would create a better user experience because people can monitor it through various devices including those that assist individuals who are disabled and those who need to split their attention between tasks. Users can utilize their hands while giving commands to the feeder using voice commands such as "Get some food to the pet now" and "How much unpacked food is left?" This enables effortless operation of the system.

○ **The Application's functionality widens to monitor health and nutritional information of pets.**

Most pet owners need methods to control their pets' nutritional health because it stands as their primary pet-related worry after getting a pet. Users can record their pet's dietary information together with how much food the pet receives and exactly when it is fed through the application programming interface. This health information center can evolve into one system through features such as weight tracking and health status updates along with scheduled veterinarian appointments. The accumulated data allows

experts to modify feeding plans and food portions for maintaining an appropriate feeding balance. The company should develop customized nutritional recommendations that consider various aspects such as pet species type and breed and age and medical situations.

A pet feeder will become an advanced pet solution which provides both feeding functionality and health-checking services along with increased interaction opportunities for owners.

9. PROBLEM SOLVING

Development of smart pet feeders demanded resolving various technological hurdles. The development process involved three technical issues alongside applicable solutions.

1. Insufficient Power Supply for Servo Motors

Problem:

The 3.3V supply from the NodeMCU system created operational difficulties for the two servo motors to perform efficiently. When supplied with this power level the motors delivered less power than designed.

Solution:

A breadboard power supply module needed to be implemented to find a solution to the problem. The breadboard power supply module delivered stable 5V power to the servo motors therefore they operated efficiently. The external power supply gave the servo motors the capabilities to work efficiently.

2. Inadequate Power for the Water Motor Problem:

The current power supply failed to initiate the water motor effectively. The insufficient power caused the motor to have chaotic startup behavior or made its operation unstable.

Solution:

We solved the issue by integrating a 9V battery as well as a relay module. The relay module worked as an electricity conductor to deliver suitable voltage for powering the water motor efficiently. The team designed a power control switch to permit effective motor operation.

3. RTC Module Detection Issue Problem:

The NodeMCU produced an error message during RTC (Real-Time Clock) module operation which read "Cannot find the RTC" through the serial monitor. The defective implementation of the RTC module failed to enable proper system configuration so the module integration was halted.

Solution:

The marketplace right now lacks an authorized solution for this particular issue. Testing for RTC module integration as well as assessing this problem will operate as simultaneous processes during implementation assessment. Attaining problem resolution demands users to check wire connections alongside confirming that NodeMCU operates with the RTC library before confirming protocol transmission.

The breakdown of problems together with proposed solutions serves as an organized system within the report explanation. Any required modifications to the solution can be implemented by me.

10. REFERENCES

Copes, F. (n.d.). *Breadboard Power Supply Module*. Flaviocopes.com.

<https://flaviocopes.com/electronics-component-power-supply/>

Schoeffler, M. (2017, November 10). *Seven Ways To Supply Power to a Breadboard - Michael Schoeffler*. Michael Schoeffler.

<https://mschoeffler.com/2017/11/10/seven-ways-to-supply-power-to-a-breadboard/>

Instructables. (2016, November 19). *IoT Pet Feeder*. Instructables.

<https://www.instructables.com/IoT-Pet-Feeder/>

11. GANTT CHART

Progress / Task	DECEMBER (01-31)							JANUARY (01-31)							FEBRUARY (01-10)						
	M	Tu	W	Th	F	Sa	Su	M	Tu	W	Th	F	Sa	Su	M	Tu	W	Th	F	Sa	Su
Proposal Preparation																					
Proposal Submission																					
Planning & Requirement Analysis																					
Hardware Setup																					
Firebase Database Setup																					
Flutter App UI/UX Design																					
Backend Development (Firebase)																					
Device Control and Implementation																					
Testing																					
Unit Testing																					
System Testing																					
Documentation																					
Final Presentation																					

12. CONCLUSION

Through the Smart Pet Feeder project developers created an advanced system which automatically feeds pets according to set schedules when owners are absent from their homes. Through implementation of advanced technology such as the real-time control system Firebase the system provides both reliability and user convenience. This smart feeding device improves homeowners' convenience while establishing foundational elements for next-generation potential that includes pet health tracking. Future modifications will transform the feeder into a full pet care system which supports pet health through a versatile solution capable of accommodating increasing pet owner needs. Technology development opens new possibilities which transform conventional pet feeding systems toward becoming essential components for contemporary pet management.